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 ✓

 ✓









Next >

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3. 3x3 Example

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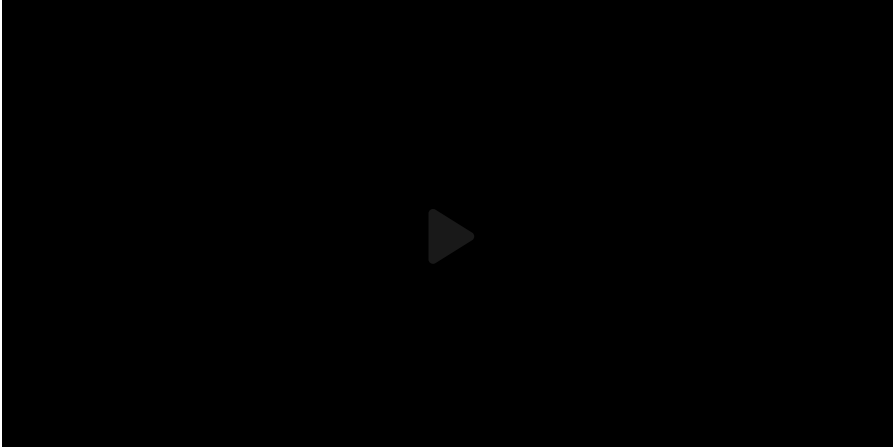
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Customizing plots in MATLAB



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MATLAB contains a variety of functions for creating common visualizations, along with customizable options for appearance and annotation to help make them clearer to others.

So, how do we customize plots in MATLAB?

Let's start by looking at a line plot of average daily

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3x3 System

Let's now revisit the case where we have three identical tanks. This was described by the following

$$\begin{aligned}\frac{dh_1}{dt} &= a_{12}(h_2 - h_1) + a_{13}(h_3 - h_1), \\ \frac{dh_2}{dt} &= a_{23}(h_3 - h_2) + a_{12}(h_1 - h_2), \\ \frac{dh_3}{dt} &= a_{23}(h_2 - h_3) + a_{13}(h_1 - h_3).\end{aligned}$$

We can then recast this in matrix form as

$$\dot{\mathbf{x}} = \begin{bmatrix} -a_{12} - a_{13} & a_{12} & a_{13} \\ a_{12} & -a_{12} - a_{23} & a_{23} \\ a_{13} & a_{23} & -a_{13} - a_{23} \end{bmatrix} \mathbf{x}$$

where $\mathbf{x} = \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$. Let's firstly set $a_{12} = a_{13} = a_{23} = 1$, and use the initial conditions

$$\mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

We know that the solution can be written in the form

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2 + c_3 e^{\lambda_3 t} \mathbf{v}_3.$$

The c_i, λ_i and $\mathbf{v}_i$ are a lot easier to find using MATLAB. Use the template below to construct the an: above system of equations using MATLAB, then plot the solution in the range $0 < t < 4$.

```
1 % Input the 3x3 matrix describing your linear system as a variable A
2 A = [-2 1 1; 1 -2 1; 1 1 -2];
3 % Input your initial conditions as a column vector x0
4 x0 = [0;1;0];
5 % Use eig(A) to find the eigenvalues and eigenvectors of A
6 % Define the eigenvectors as column vectors v1, v2 and v3,
7 % and the eigenvalues as lambda1, lambda2, lambda3
8 % so that A*v1 = lambda1*v1, etc.
9 [V,D] = eig(A);
10 v1 = V(:,1);
11 v2 = V(:,2);
12 v3 = V(:,3);
13 lambda1 = D(1,1);
14 lambda2 = D(2,2);
15 lambda3 = D(3,3);
16 % Calculate the column vector c = [c1;c2;c3] from the initial conditions using
17 c = inv(V)*x0;
18 c1 = c(1,1); c2 = c(2,1); c3 = c(3,1);
19 %%% Define a row vector t with 100 equally spaced entries,
20 %%% beginning with 0 and ending at 4.
21 t = linspace(0,4,100);
22 %%% Define three row vectors h1, h2 and h3, with entries corresponding
23 %%% to h1(t), h2(t) and h3(t) evaluated at each time in t.
24 h1 = c1*exp(lambda1*t)*v1(1) + c2*exp(lambda2*t)*v2(1) + c3*exp(lambda3*t)*v3(1);
25 h2 = c1*exp(lambda1*t)*v1(2) + c2*exp(lambda2*t)*v2(2) + c3*exp(lambda3*t)*v3(2);
26 h3 = c1*exp(lambda1*t)*v1(3) + c2*exp(lambda2*t)*v2(3) + c3*exp(lambda3*t)*v3(3);
27 %%% Now use plot to plot the three vectors against time on the same figure
28 plot(t,h1) %tell MATLAB to plot h1 here
29 %
30 hold on % You need to put this command in between the plot commands so that
31 %
32 plot(t,h2) %tell MATLAB to plot h2 here
33 %
34 hold on % You need to put this command in between the plot commands so that
35 %
36 plot(t,h3) %tell MATLAB to plot h2 here
37
38 set(gca,'fontsize',18)
39 xlabel('Time')
40 ylabel('Volume')
41 title('Time series')
42
```

 Run Script



Assessment: All Tests Passed

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 h1

 h2

 h3

 plot used

3. 3x3 Example

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